$0 \simeq (1 \diamond 0) \diamond 2$. We then replace every equational law with their minimal element in their definitional equivalence class, which can be viewed as the *normal form* for that law; for instance, the normal form of $(0 \diamond 1) \diamond 2 \simeq 1$ would be $0 \simeq (1 \diamond 0) \diamond 2$. Finally, we eliminate any law of the form $w \simeq w$ other than $0 \simeq 0$. We then number the remaining equations $\mathrm{E1}, \mathrm{E2}, \ldots$. For instance, E1 is the trivial law $0 \simeq 0$, E2 is the singleton law $0 \simeq 1$, E3 is the idempotent law $0 \simeq 0 \diamond 0$, and so forth. Lists and code for generating these equations, or the equation number attached to a given equation, can be found in the ETP repository.

The number of equations in this list of order $n = 0, 1, 2, \ldots$ is given by

$$2, 5, 39, 364, 4284, 57882, 888365, \ldots$$

(`https://oeis.org/A376640`). The number can be computed to be

$$C_{n+1} B_{n+2}/2$$

if n is odd, 2 if $n = 0$, and

$$(C_{n+1} B_{n+2} + C_{n/2}(2D_{n+2} - B_{n+2}))/2 - C_{n/2} B_{n/2+1}$$

if $n > 2$ is even, where C_n, B_n are the Catalan and Bell numbers, and D_n is the number of partitions of $[n]$ up to reflection, which for $n = 0, 1, 2, \ldots$ is

$$1, 1, 2, 4, 11, 32, 117, \ldots$$

(`https://oeis.org/A103293`). A proof of this claim can be found in the ETP blueprint. In particular, there are 4694 equations of order at most 4.

Below we record some specific equations appearing in this paper, using the alphabet x, y, z, w in place of 0, 1, 2, 3 for readability.

(E1)	$\mathrm{x} \simeq \mathrm{x}$	(Trivial law)
(E2)	$\mathrm{x} \simeq \mathrm{y}$	(Singleton law)
(E3)	$\mathrm{x} \simeq \mathrm{x} \diamond \mathrm{x}$	(Idempotent law)
(E4)	$\mathrm{x} \simeq \mathrm{x} \diamond \mathrm{y}$	(Left-absorptive law)
(E5)	$\mathrm{x} \simeq \mathrm{y} \diamond \mathrm{x}$	(Right-absorptive law)
(E10)	$\mathrm{x} \simeq \mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{x})$	
(E11)	$\mathrm{x} \simeq \mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{y})$	(Right-unit squares law)
(E14)	$\mathrm{x} \simeq \mathrm{y} \diamond (\mathrm{x} \diamond \mathrm{y})$	(Semi-symmetric quasigroups)
(E23)	$\mathrm{x} \simeq (\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{x}$	
(E40)	$\mathrm{x} \diamond \mathrm{x} \simeq \mathrm{y} \diamond \mathrm{y}$	(Unipotence law)
(E41)	$\mathrm{x} \diamond \mathrm{x} \simeq \mathrm{y} \diamond \mathrm{z}$	(Equivalent to (E46))
(E43)	$\mathrm{x} \diamond \mathrm{y} \simeq \mathrm{y} \diamond \mathrm{x}$	(Commutative law)
(E46)	$\mathrm{x} \diamond \mathrm{y} \simeq \mathrm{z} \diamond \mathrm{w}$	(Constant law)
(E47)	$\mathrm{x} \simeq \mathrm{x} \diamond (\mathrm{x} \diamond (\mathrm{x} \diamond \mathrm{x}))$	
(E73)	$\mathrm{x} \simeq \mathrm{y} \diamond (\mathrm{y} \diamond (\mathrm{x} \diamond \mathrm{y}))$	
(E151)	$\mathrm{x} \simeq (\mathrm{x} \diamond \mathrm{x}) \diamond (\mathrm{x} \diamond \mathrm{x})$	
(E168)	$\mathrm{x} \simeq (\mathrm{y} \diamond \mathrm{x}) \diamond (\mathrm{x} \diamond \mathrm{z})$	(Central groupoid law)

(E206)	$x \simeq (x \diamond (x \diamond y)) \diamond y$	
(E255)	$x \simeq ((x \diamond x) \diamond x) \diamond x$	
(E327)	$x \diamond y \simeq x \diamond (y \diamond z)$	(Right reduction law)
(E378)	$x \diamond y \simeq (x \diamond y) \diamond y$	(Right idempotence law)
(E395)	$x \diamond y \simeq (z \diamond x) \diamond y$	(Left reduction law)
(E413)	$x \simeq x \diamond (x \diamond (x \diamond (y \diamond x)))$	
(E450)	$x \simeq x \diamond (y \diamond (z \diamond (y \diamond x)))$	
(E492)	$x \simeq y \diamond (x \diamond (z \diamond (z \diamond y)))$	(Totally symmetric quasigroups)
(E543)	$x \simeq y \diamond (z \diamond (x \diamond (y \diamond z)))$	(Tarski's axiom)
(E650)	$x \simeq x \diamond (y \diamond ((z \diamond x) \diamond y))$	(Non-trivially equivalent to (E4))
(E677)	$x \simeq y \diamond (x \diamond ((y \diamond x) \diamond y))$	(Last open implication)
(E817)	$x \simeq x \diamond ((x \diamond x) \diamond (x \diamond x))$	
(E854)	$x \simeq x \diamond ((y \diamond z) \diamond (x \diamond z))$	
(E1045)	$x \simeq x \diamond ((y \diamond (y \diamond x)) \diamond x)$	
(E1055)	$x \simeq x \diamond ((y \diamond (z \diamond x)) \diamond x)$	
(E1110)	$x \simeq y \diamond ((y \diamond (x \diamond x)) \diamond y)$	
(E1117)	$x \simeq y \diamond ((y \diamond (x \diamond z)) \diamond z)$	
(E1286)	$x \simeq y \diamond (((x \diamond y) \diamond x) \diamond y)$	
(E1323)	$x \simeq y \diamond (((y \diamond y) \diamond x) \diamond y)$	
(E1485)	$x \simeq (y \diamond x) \diamond (x \diamond (z \diamond y))$	(Weak central groupoids)
(E1518)	$x \simeq (y \diamond y) \diamond (x \diamond (y \diamond x))$	
(E1571)	$x \simeq (y \diamond z) \diamond (y \diamond (x \diamond z))$	(Boolean groups)
(E1629)	$x \simeq (x \diamond x) \diamond ((x \diamond x) \diamond x)$	
(E1648)	$x \simeq (x \diamond y) \diamond ((x \diamond y) \diamond y)$	
(E1659)	$x \simeq (x \diamond y) \diamond ((y \diamond y) \diamond z)$	
(E1689)	$x \simeq (y \diamond x) \diamond ((x \diamond z) \diamond z)$	(Equivalent to (E2))
(E1729)	$x \simeq (y \diamond y) \diamond ((y \diamond x) \diamond y)$	
(E2301)	$x \simeq (y \diamond (x \diamond (y \diamond x))) \diamond y$	
(E2441)	$x \simeq (x \diamond ((x \diamond x) \diamond x)) \diamond x$	
(E2744)	$x \simeq ((y \diamond y) \diamond (y \diamond x)) \diamond y$	
(E2910)	$x \simeq ((y \diamond (x \diamond y)) \diamond x) \diamond y$	(Dual of (E677))
(E3316)	$x \diamond y \simeq x \diamond (y \diamond (x \diamond y))$	
(E3523)	$x \diamond y \simeq x \diamond ((y \diamond y) \diamond z)$	
(E3737)	$x \diamond y \simeq (x \diamond z) \diamond (y \diamond z)$	
(E3925)	$x \diamond y \simeq (x \diamond (y \diamond x)) \diamond y$	
(E4315)	$x \diamond (y \diamond x) \simeq x \diamond (y \diamond z)$	

(E4380)	$x \diamond (x \diamond x) \simeq (x \diamond x) \diamond x$	(Cube-associativity law)
(E4482)	$x \diamond (y \diamond y) \simeq (y \diamond y) \diamond x$	(Central squares law)
(E4512)	$x \diamond (y \diamond z) \simeq (x \diamond y) \diamond z$	(Associative law)
(E4531)	$x \diamond (y \diamond z) \simeq (y \diamond z) \diamond x$	(Central products law)
(E5093)	$x \simeq y \diamond (y \diamond (y \diamond (x \diamond (z \diamond y))))$	
(E85914)	$x \simeq y \diamond (z \diamond ((y \diamond x) \diamond (z \diamond (y \diamond z))))$	(Eisenstein modules)
(E86082)	$x \simeq y \diamond (z \diamond ((y \diamond z) \diamond (w \diamond (x \diamond w))))$	(Gaussian modules)
(E345169)	$x \simeq (y \diamond ((x \diamond y) \diamond y)) \diamond (x \diamond (z \diamond y))$	(Sheffer stroke)

We also list some order-8 characterizations of group division relevant for Section 10.

(E42302852)	$x \simeq y \diamond ((((x \diamond x) \diamond x) \diamond z) \diamond (((x \diamond x) \diamond y) \diamond z))$	(McCune law)
(E42302946)	$x \simeq y \diamond ((((x \diamond x) \diamond x) \diamond z) \diamond (((y \diamond y) \diamond y) \diamond z))$	
(E42323216)	$x \simeq y \diamond ((((y \diamond y) \diamond x) \diamond z) \diamond (((y \diamond y) \diamond y) \diamond z))$	(Higman–Neumann law)
(E67953597)	$x \simeq (y \diamond y) \diamond (y \diamond ((x \diamond z) \diamond (((x \diamond x) \diamond y) \diamond z)))$	(in finite magmas)
(E89176740)	$x \simeq ((y \diamond y) \diamond y) \diamond ((((x \diamond x) \diamond x) \diamond z) \diamond (y \diamond z))$	
(E102744082)	$x \simeq ((y \diamond y) \diamond ((x \diamond z) \diamond x)) \diamond ((z \diamond w) \diamond (x \diamond w))$	
(E147976245)	$x \simeq ((y \diamond y) \diamond (y \diamond (x \diamond (((y \diamond y) \diamond y) \diamond z)))) \diamond z$	(McCune law)

Appendix B. Author contributions

In a companion document to this paper, the contributions of each author of this paper to the ETP are described, following the standard CRediT categories[41]. Below are the affiliations and grant acknowledgments of individual participants.

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[41] `https://credit.niso.org/`